Figure 16-20 Materials for a soft padded bandage (modified Robert Jones bandage). **A,** Materials needed to start the bandage and materials for the contact layer. **B** and **C,** Materials for the secondary layer of bandage (padding and compression). **D,** Materials for the tertiary or outer layer of bandage.

over a pressure point, such as the tuber calcis, olecranon, or accessory carpal bone, to prevent pressure sores. The doughnut can be made from cotton cast padding or roll cotton. After folding several layers a hole may be cut in the center so that the doughnut fits over the pressure point. The purpose of the doughnut is to distribute the forces of bandage compression over a greater surface area rather than just over the pressure point. Just adding more cotton over a pressure point without a doughnut may actually increase the pressure over the point.

When beginning application of the bandage, cast padding or roll cotton is placed circumferentially around the limb, always rolling the material from distal to proximal (Figure 16-21, *A*). The tips of the middle digits should be barely showing from the end of the bandage. This will allow veterinary care providers and owners to check for coolness and swelling of the digits, which might indicate that the bandage is too tight. When applying material it is helpful to roll the material on from lateral to medial, overlapping by 50% and distributing pressure uniformly throughout all layers. Cast padding is often textured and the textured markings flatten when the material is pulled taut. If too much tension is applied the material will tear. If the secondary layer of cast padding is too loose the bandage may slip distally and irritate the tissues under the bandage. A compression layer is then applied over the cotton, using gauze material applied under tension while wrapping the gauze circumferentially from distal to proximal, and applying it in such a fashion as to cause slight internal rotation of the limb (Figure 16-21, *B*). The gauze should not be placed beyond the cotton padding, which could cause vascular compromise. It is very easy to apply the rolled gauze too tight, so careful application is necessary. When working on the proximal aspects of the bandage, it might be helpful to have the limb abducted to allow the cast padding, rolled gauze, and Vet Wrap to be applied as proximally as possible. When applying the tertiary outer layer the material should be loosely applied by either unwrapping it and then carefully re-wrapping it to loosen the elasticity of the material and ensure the material is

Figure 16-21 Placement of a soft padded (modified Robert Jones) bandage. **A,** Application of rolled cotton. **B,** Application of rolled gauze to provide compression. **C** and **D,** Application of the outer layer. Note: all layers are applied in a distal to proximal direction.

not applied too tight, or unroll some of the material, allow the material to undergo elastic recoil, and then roll it on the limb as it is applied in a distal to proximal direction (Figure 16-21, *C* and *D*). Some practitioners apply a strip of Elastikon at the most proximal aspect of the bandage to the skin to help prevent bandage slipping. Care must be taken not to apply adhesive tape over thin-skinned areas, such as the medial thigh, to avoid irritation of the skin by the adhesive.

Soft-Padded Bandages

The Robert Jones bandage is one of the most frequently applied coaptation bandages in veterinary medicine. It provides excellent temporary support for an injured limb before or after surgery. Overall the Robert Jones bandage is a safe bandage and few complications are seen. The main layer is rolled cotton, which yields the final diameter of the bandage, which is 40% to 50% larger than the original limb. The modified Robert Jones bandage is similar to

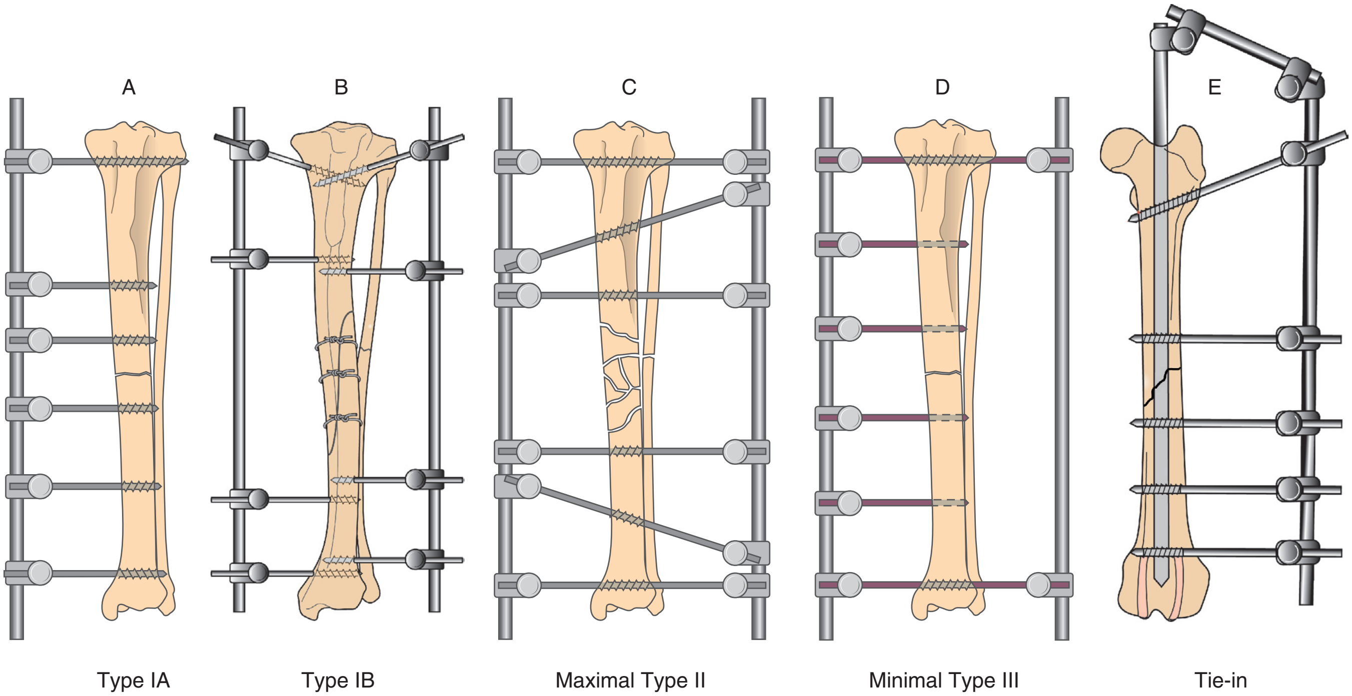

Figure 16-22 Classification of external fixator frame configurations. **A,** Type IA. One plane, unilateral. **B,** Type IB. Two plane, unilateral. **C,** Type II. One plane, bilateral. **D,** Type III. Two plane, bilateral. **E,** Tie-in configuration. (From Fossum, TW: *Small Animal Surgery*, ed. 4, St Louis, 2013, Mosby. Second and fourth images modified from Johnson AL, Dunning D: *Atlas of orthopedic surgical procedures of the dog and cat*, St. Louis, 2005, Saunders in Fossum, TW: *Small Animal Surgery,* ed. 4, St Louis, 2013, Mosby.)

the Robert Jones, except less cotton layer is applied, thus significantly decreasing the size of the bandage. The modified Robert Jones bandage provides the same advantages as the Robert Jones bandage, yet it is more economical and tolerable for the patient. The modified Robert Jones bandage is the most commonly applied bandage following surgeries of the elbow or stifle, or distal to these joints. However, neither the Robert Jones bandage nor the modified Robert Jones bandage provides adequate stabilization for fractures. Bandages should be changed daily up to weekly as directed by the veterinary surgeon.

Bandaging an External Fixator

External fixators are used to repair fractures or joint luxations and may be applied in various configurations. They can counteract the forces of rotation, bending, shearing, and distraction. External fixator configurations can be designed to meet the stabilization needs of a fracture and can be subsequently modified or destabilized throughout the healing process.[21] Pin factors, such as pin type, size, number, location and length, and frame configuration, affect the stiffness of the fixator and its ability to resist the forces associated with weight bearing, such as bending, axial loading, and rotation.

There are three main types of linear external fixators (Figure 16-22). The fixator configurations are named for the number of planes occupied by the fixator and the number of sides of the limb from which the fixator protrudes.

- Type 1A is unilateral-uniplanar; usually applied to the medial aspect of the radius or tibia and to the lateral humerus or femur.
- Type 1B is unilateral-biplanar; usually applied to the radius or tibia.
- Type 2 is bilateral-uniplanar; applied only to the radius or tibia, it cannot be applied to the humerus or femur because of the close proximity to the body on the medial side.
- Type 3 is bilateral-biplanar; applied only to the radius or tibia, it cannot be applied to the humerus or femur because of the close proximity to the body on the medial side.

The number of pins and planes directly affect the strength and stiffness of the external fixator. When an external fixator is used with an IM pin, it is called a tie-in configuration, and results in greater strength and stiffness of the fixation.

Pins may be placed in different ways. Half pins penetrate both cortices of the bone but only one skin surface. Full pins penetrate both cortices and both skin surfaces. Clamps and connecting rods adjoin fixator pins and connecting bars to each other.

Circular or Ilizarov ring external fixators are multiplanar and multilateral, facilitate distraction of long bones, and may be used to correct angular limb deformities.

Special wires or small pins are placed through the bone and connected to rings with clamps. Distraction osteogenesis induces new bone formation between two bone ends (after an osteotomy is performed) that are distracted over time. The rate of distraction osteogenesis is 1 mm per day and the rhythm is 0.25 mm four times a day.[21]

External fixators provide a bandaging challenge. Although there are many different methods to bandage an external fixator, certain principles should be followed. Each pin-skin interface should be covered with a primary contact layer. Often a sterile nonadherent pad is cut to accommodate the pin and trimmed to an appropriate smaller size. Sterile triple antibiotic ointment is placed on each pad before applying the contact layer to the patient (Figure 16-23, *A*). The pads can be held in place with recycled dry scrub sponges placed around the pins, connecting rods, and linkage devices (Figure 16-23, *B*). The sponges also provide padding for the patient and the apparatus. All sharp pins protruding from the apparatus should be covered to protect the patient. The ability to improvise is important; items used to cover pin points include rubber caps, injection caps, and sponges folded over the pin point and securely taped (Figure 16-23, *C*). Next the apparatus is wrapped with cast padding, roll gauze, and Vet Wrap, noting that the limb itself is *not* incorporated in the bandage (Figure 16-23, *D* through *I*) unless edema is present or wounds must be covered. In these cases, the bandage is placed similar to a modified Robert Jones bandage, incorporating the external fixator as previously described. An adequate but minimum amount of cast padding should be applied because if the bandage is too large it may slip off.

External fixator bandages should be rechecked weekly. The bandage is removed, the apparatus is assessed for loosening or drainage, and radiographs are taken at the veterinarian's discretion. While the patient is still sedated from the radiographs, the pin tracts are thoroughly cleaned

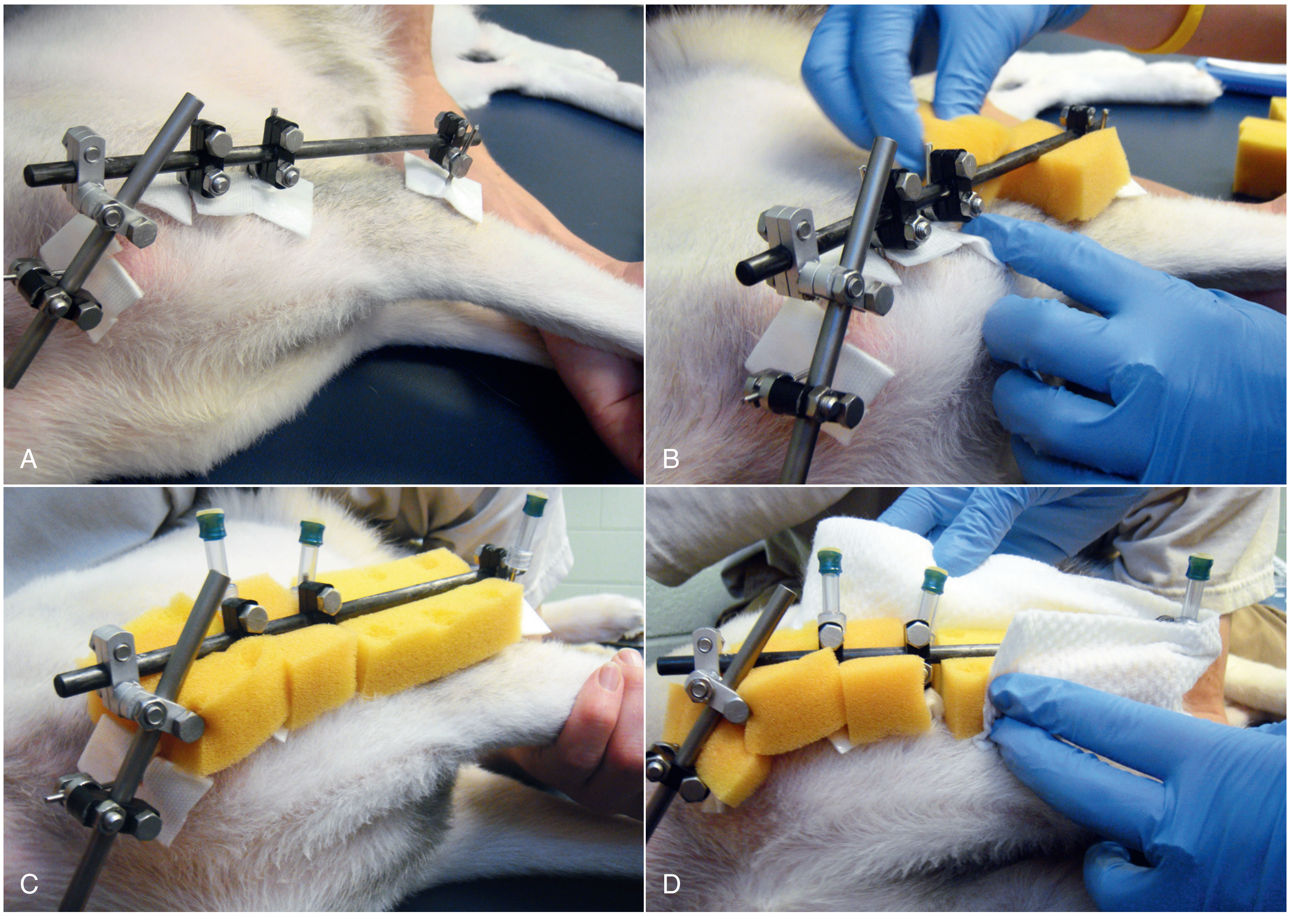

Figure 16-23 Bandage placement over an external skeletal fixator. **A,** Nonadherent pads cut to accommodate external fixator pins for treatment of a fracture of the humerus. **B,** Recycled scrub sponges, cleaned and dried, can be used to hold the nonadherent pads in place and provide padding. **C,** Injection caps can be used to cover pin points to protect the patient and environment. **D** and **E,** Cotton cast padding is applied to the apparatus.

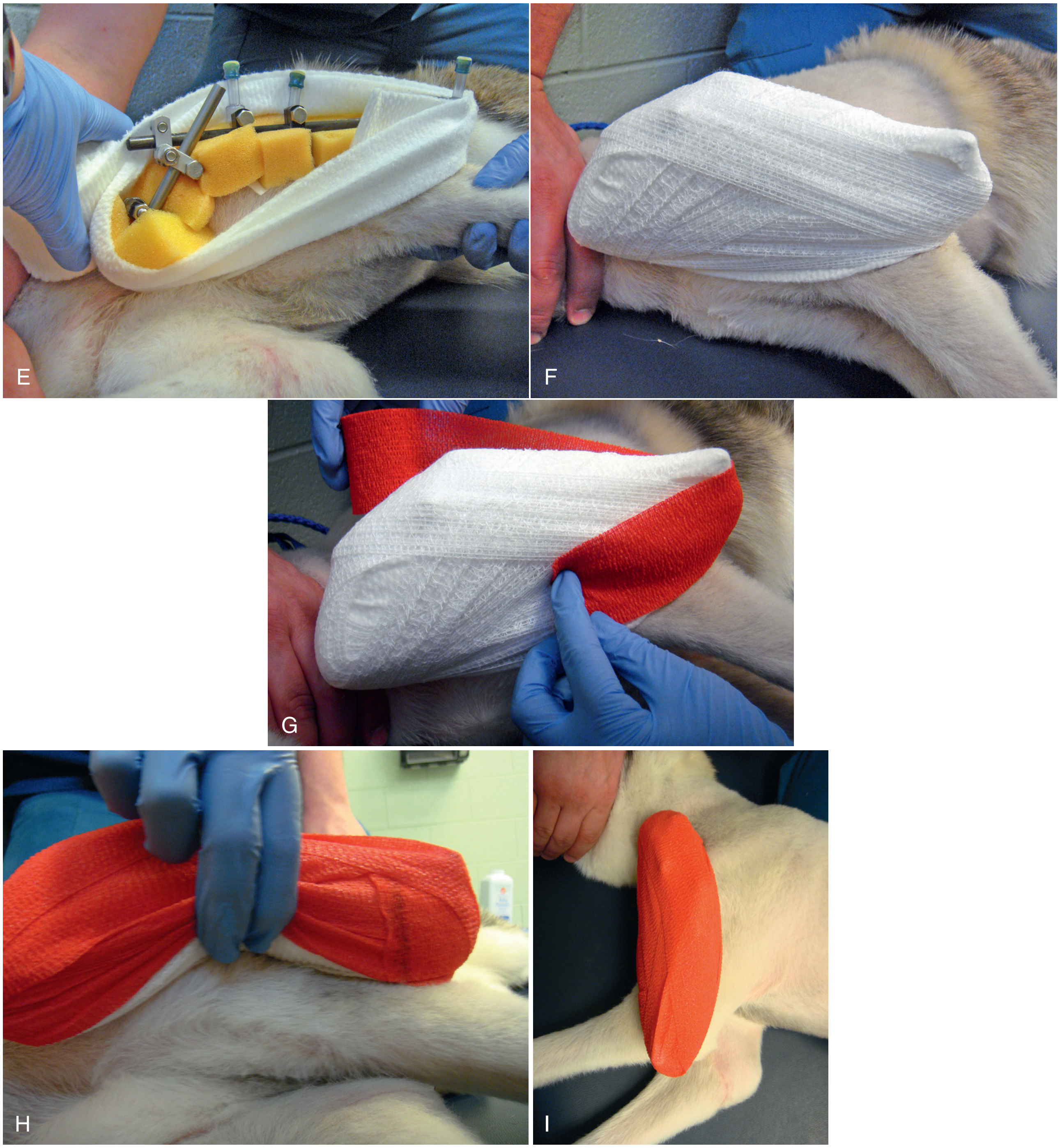

Figure 16-23, cont'd **F,** Roll gauze is applied in a similar manner. **G,** An outer wrap is applied in a similar manner. **H,** Note that the limb is not included in the bandage, unless edema is present or wounds must be covered. **I,** Finished product: bandage placed over an external skeletal fixator of the humerus.

and supplies are prepared for bandage change (Figure 16-24). Using a cotton ball or gauze sponge and chlorhexidine scrub, gently scrub around the pin tracts to remove dried exudate; sterile cotton tipped applicators also may be used. Gently rinse with warm saline on a cotton ball, gauze, or cotton tipped applicators. This process is repeated for each pin tract until all are free of dried exudates (Figure 16-25). When accessible it may be necessary to scrub and rinse portions of the limb. The limb is allowed to air dry or a hair dryer placed on the cool and low setting can be used to expedite the drying process as tolerated by the patient. Depending on radiographic healing the veterinarian may choose to remove some pins, exchange connecting rods, or remove the device entirely. Removing the fixator

Figure 16-24 Wound management and bandaging materials needed for bandage change of an external fixator bandage of the humerus.

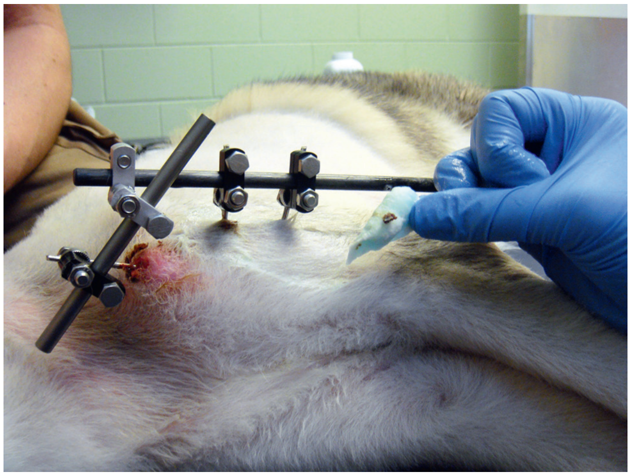

Figure 16-25 Pin tracts are cleaned with dilute chlorhexidine scrub and warm saline to remove exudate.

in stages gradually reduces the stability of the fixator and increases loading of the bone, which helps to stimulate callus development and maturation. The external fixator is rebandaged at each recheck or adjustment. The process of weekly rechecks continues as directed by the veterinarian until the injury has healed and the apparatus can be removed.

Pin loosening is the most common postoperative complication of external fixators. Factors that contribute to pin loosening are excessive micromotion, damage to the bone at the time of insertion, and fatigue failure of the bone cortex.[22] Pin tract infections can also occur and result in increasing purulent discharge from the pin-skin interface, reduced limb use, and pain with palpation over the site. In addition, delayed fracture healing, joint stiffness, and iatrogenic fracture are other complications that can occur.

Splints

A splint is essentially a bandage with incorporation of a stiff supportive device. Splints can be made from various materials. Commercial splints are available, such as spoon splints, Mason Metasplints, and Quick Splints (Jorgensen Laboratories, Loveland, CO). Some splints can be custom made using aluminum rods or casting material. Finding the appropriate size of commercially available splints for a particular patient can present a challenge. It is important that the splint fit properly. Mason Metasplints and Spoon Splints should cradle the caudal aspect of the limb and not extend proximal to the elbow or tarsus. Quick Splints should fit properly at every joint angle. It is important the splint is applied between layers of the bandage, most commonly after the compression layer has been applied.

Spica splints are used to immobilize all joints of the forelimb or hind limb, but are more commonly applied to the forelimb. The Spica splint can be applied in cases of humeral fracture repair, and closed or open reduction of an elbow luxation. The Spica splint envelops the affected limb and the torso of the patient. The limb is held in a functional standing position. After cast padding has been applied, the splint material, such as premeasured and prebent rods or fiberglass cast material, is applied from the distal digits over the dorsum to the proximal scapula of the contralateral limb. Roll gauze is then applied followed by an outer wrap. Monitoring the splint after application is important because Spica splints can cause abrasions in the axillary area.

Casts

Casts are a conservative method to stabilize fractures in young patients after closed reduction. Common fractures include simple fractures of the radius and ulna or fractures of the radius with the ulna intact. Similarly simple fractures of the tibia and fibula or fractures of the tibia with the fibula intact may be managed with a cast.

Cast application involves making sure the leg is clean and dry before application. Long hair should be clipped to a shorter length before cast placement, but not surgically clipped or shaved. To maintain patient comfort during reduction of the fracture and application of the cast, the patient is lightly anesthetized or heavily sedated with adequate analgesics. Tape stirrups are applied to either the medial and lateral or cranial and caudal aspects of the distal limb. An appropriate sized stockinette is applied to the leg. Traction is then placed on the limb to aid in fracture reduction. The fracture site may be palpated and manipulated to aid in reduction. It is important that the limb is not placed such that varus or valgus deformation of the limb is present. To avoid these deformations the limb may be placed so that the limb is hung from the ceiling during fracture reduction to make alignment of the limb easier.

Circular pads or doughnuts are applied over bony prominences. Cast padding is then applied, moving in a spiral

Figure 16-26 To maintain coverage and protection of a bandage, commercially made bandage covers can be used.

pattern from distal to proximal, overlapping by 50%, with up to two to four layers applied. Fiberglass cast material is immersed in tepid water (70-75° F). The warmer the water, the faster the cast will harden. When the cast material is immersed bubbles will be released. After bubbling has ceased (10-15 seconds), the material can be removed from the water and shaken, but not squeezed, to remove excess water. A second person maintains the limb in traction with the fracture reduced while the cast is applied. After making a final assessment of the limb for proper alignment, the cast material is applied in a distal to proximal spiral pattern, overlapping by 50%. After the first layer is applied the tape stirrups and stockinette are turned up to incorporate them into the cast. Continue to apply cast material so that two to four layers are applied. Gently laminate the layers together and allow for adequate curing time. The cast material undergoes an exothermic reaction that creates heat during the curing process.

A saw may be used to cut the cast on the medial and lateral aspects of the limb after curing to bivalve the cast. When cutting the cast material with the saw, it is important to hold the saw stationary as it cuts. Moving the saw back and forth while cutting may result in cutting the patient's skin. Bivalving of the cast facilitates regular rechecks and assessment of the limb underneath. The patient may weight bear on a casted limb as early as 20 minutes after application is complete when using fiberglass casting material.

Maintenance of the External Coaptation Device

It is very important to keep all bandages, splints, slings, and casts clean, dry and intact. The external coaptation device must be protected from adverse weather. Plastic bags, recycled and dried fluid bags, or commercially made bandage covers (e.g., Medipaw, MediVet Products, Fogelsville, PA) may be used to cover the distal end of a bandage, splint, sling, or cast when walking outside (Figure 16-26). The patient should be housed inside while wearing an external coaptation device and the plastic covering should be removed when inside to prevent moisture buildup inside the bandage. Bandages should be checked several times daily for swollen toes, slipping of the bandage, moisture, soiling, drainage, or foul odor. The external coaptation device should be changed immediately if any of these is present, or if the device is damaged or loosened. If the bandage has been well tolerated and the patient begins to lick or chew at the bandage or toes, the bandage should be changed because there is generally a problem under the bandage that should be addressed as soon as possible, such as a wound infection, moisture that is irritating the skin, or wound dehiscence.

Figure 16-27 The Velpeau sling maintains the carpus, elbow and shoulder in a flexed, non–weight-bearing position.

Complications of Bandaging

The complications of bandaging include pressure sores and ischemic injury, infection, damage to the surrounding soft tissues, neuropraxia, contracture, malunion, and angular deformities. The complications from a bandage can be serious. Therefore proper bandage application and aftercare cannot be overstated.

Slings

A sling is designed to prevent the patient from weight bearing on the affected limb. A Velpeau sling maintains the carpus, elbow, and shoulder in a flexed, non–weight-bearing position (Figure 16-27). The Velpeau sling can be applied in cases of scapular body fracture, bicipital tenosynovitis, and medial shoulder luxation. To apply a Velpeau sling: Flex the shoulder and elbow while adducting the limb against the body; start by applying two or three layers of cast padding around the patient's torso and affected forelimb, incorporating the forelimb such that it is secured to the thoracic wall. Some layers should go in a cranial to caudal direction around the limb to prevent slippage of the incorporated limb. Apply rolled gauze in the same manner to provide mild compression. Finally apply the outer wrap in the same manner to provide support.

An Ehmer sling maintains the hock, stifle, and hip in flexion, with internal rotation of the hip (Figure 16-28). The Ehmer is commonly used following closed or open

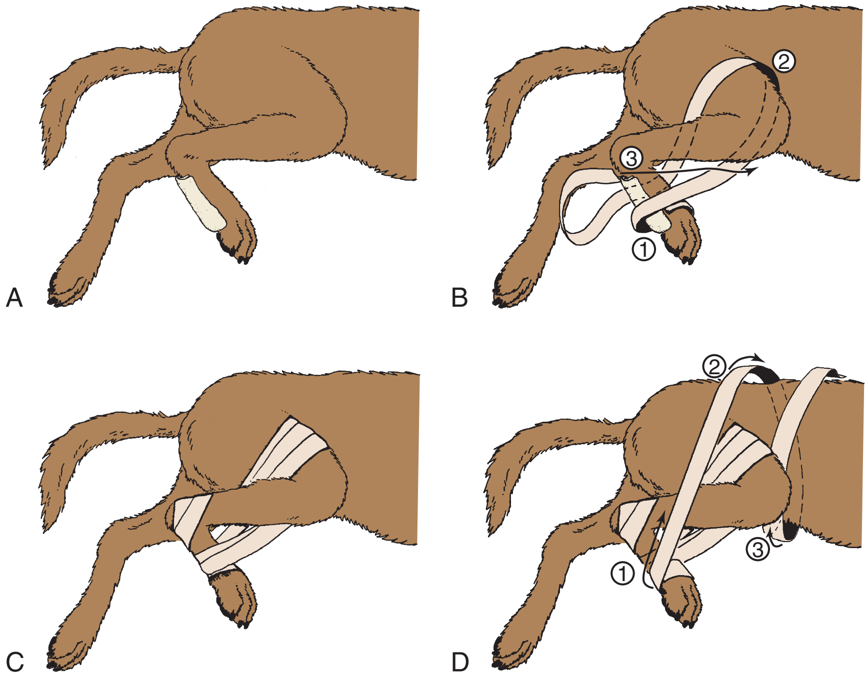

Figure 16-28 Ehmer (figure-of-8) sling. **A,** Application of the sling begins with placing some padding on the plantar surface of the metatarsus. **B,** Conforming roller gauze bandage is wrapped around the metatarsus (1) from lateral to medial, being sure to include most of the metatarsal pad. After several wraps to secure it, the gauze is carried medial to the flexed stifle (2) and over the cranial surface of the thigh. This internally rotates the limb at the hip joint. Finally (3) the gauze is brought medial to the tibia and tarsus and over the plantar surface of the tarsus. Several more circuits are made in the same manner. **C,** The gauze is continued in a figure-of-8 fashion around the flexed hock and paw to secure it. The entire bandage is then covered with elastic adhesive tape that overhangs the gauze to anchor the sling to the hair. Although some construct the bandage without any gauze by starting with adhesive tape applied to the skin, this invariably leads to considerable skin irritation on the cranial thigh region as well as the difficulty of removing the large amount of tape. **D,** It is difficult to keep the sling from slipping down over the stifle on short-legged breeds. One solution is to attach wide adhesive tape to the paw area of the completed sling (1), then carry the tape upward over the back (2) and around the belly (3). This is simple in the bitch or cat, but care must be taken to avoid the sheath in the male dog. (Adapted from Piermattei, DL, Flo GL, DeCamp CE: *Handbook of small animal orthopedics and fracture repair,* ed 4, St. Louis, 2006, Saunders.)

reduction of craniodorsal hip luxations, as well as some acetabular fractures, and can be maintained for up to two weeks. To apply an Ehmer sling, place several layers of cast padding around the metatarsal region. Next wrap roll gauze over it several times. Fully flex the stifle and wrap the rolled gauze around the thigh by bringing it medial to the thigh. Next, pull the gauze firmly and bring it over the front of the thigh to maintain flexion. Wrap the gauze over the lateral aspect of the thigh and then wrap it distally medial to the tarsus. The wrap is continued around the plantar aspect of the metatarsals, wrapping from medial to lateral to return to the original starting position. Repeat the pattern with rolled gauze three or four times. Finish with a layer of Elastikon applied in the same manner.

The pelvic limb sling, also known as a Robinson sling, prevents full weight bearing without maintaining the joints in a highly flexed position (Figure 16-29). A pelvic limb sling suspends the metatarsals to a bellyband and can be maintained for many weeks. Periodic recheck examinations are needed to maintain the integrity of the sling; replace it if the sling becomes soiled or wet. An e-collar is recommended to prevent the patient from removing the sling. A pelvic limb sling can be applied following fracture repair of the femur and tibia or in other situations where joint motion is desired without full weight bearing.

A carpal flexion sling prevents weight bearing on a forelimb and removes stress from flexor tendons (Figure 16-30). However, it is important to remember that prolonged application results in limitations in carpal extension and may require stretching and range of motion to reestablish normal extension of the carpus and digits. A carpal flexion sling can be applied following flexor tendon or fracture repair.

The 90/90 flexion sling is designed to maintain the tarsus and stifle in 90 degrees of flexion and is primarily used to prevent contracture of the quadriceps following repair of fractures of the distal femur in young dogs. To apply a 90/90 flexion sling, flex the stifle and tarsus to 90 degrees of flexion. With cast padding, wrap several layers around the metatarsal region and over the distal thigh. Adhesive tape or Elastikon is then wrapped around the flexed tarsal and stifle joints to maintain 90 degrees of flexion.

Hobbles may be applied to the distal antebrachii or cruses to allow full weight bearing and walking, but prevent

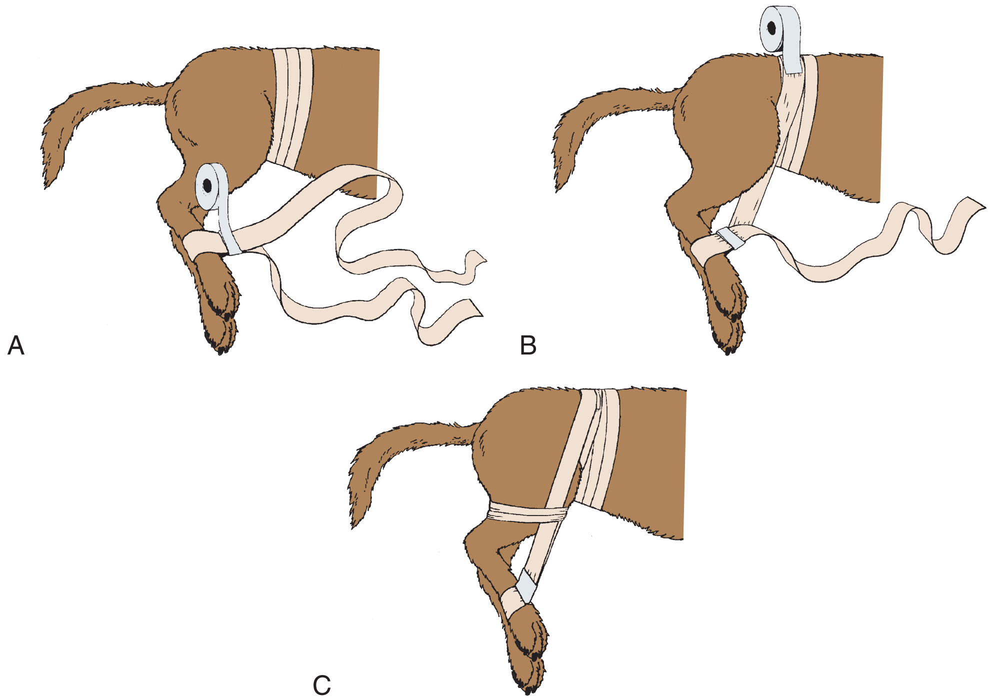

Figure 16-29 Robinson sling. **A,** The metatarsal region is padded with cotton or cast padding. A long piece of adhesive tape or Elastikon is wrapped around the metatarsal region, and the tape is stuck to itself near the foot. A belly wrap is also placed by wrapping the caudal abdomen with cast padding or roll cotton. Adhesive tape or Elastikon is then wrapped around the cotton. **B,** The medial strip of tape is then placed alongside the belly wrap, shortening the tape to an appropriate length so as to prevent weight bearing on the foot when standing. This strip is secured to the belly wrap with additional tape. **C,** The lateral strip of tape is then passed over the lateral surface of the thigh and attached to the belly wrap with additional tape. (From Piermattei, DL, Flo GL, DeCamp CE: *Handbook of small animal orthopedics and fracture repair,* ed 4, St. Louis, 2006, Saunders.)

Figure 16-30 Carpal flexion sling. This sling prevents weight bearing on a forelimb and removes stress from flexor tendons.

abduction. The most common use is in dogs with medial glenohumeral instability or ventral coxofemoral luxation. Cast padding is placed around both distal limbs. Adhesive tape may then be wrapped around the limbs to secure them in a standing position, but one which will not allow abduction to occur. More recently commercially fabricated hobbles that may be customized to the patient are available (Shoulder Stabilization System, DogLeggs, Reston, VA).

Aftercare and Complications of Slings

If care is not taken in placing slings, serious complications may arise such as tissue necrosis, neuropraxia, and severe limb edema. Toes should be checked at least twice daily for excessive swelling and warmth. If the toes are cool or swollen, the sling should be immediately removed and a new one placed after the clinician is certain that there are no circulatory problems. Other commercial alternatives have become available in recent years, using Velcro fasteners to make application easier (DogLeggs, Reston, VA). These slings are reusable and washable and allow for daily removal and reapplication for treatments such as passive range of motion exercises (Figure 16-31).

Postoperative Management of Bowel and Bladder

Patients often have decreased intestinal motility while in the hospital and thus may not defecate for days after surgery. Clients frequently express concern and want to know what they can do. Multiple factors contribute to decreased defecation, including discomfort when posturing to defecate or constipation or pain when defecating,

Figure 16-31 Removable and reusable slings are commercially available for the rear limbs. Slings are also available for the forelimbs (not shown).

such as following a pelvic fracture. Metamucil, DSS, or lactulose may be administered to soften stools. Another treatment that is simple and economical is the addition of cooked pumpkin to the food, which is high in fiber and serves as a source of insoluble fiber.

Urine present in the patient's cage is not evidence that the patient can urinate voluntarily or that the urinary system is intact after trauma. Appropriate evaluation of the urinary system should be made to be certain that it is intact after trauma. Urinary retention can lead to serious urinary tract infections, possibly even ascending to the kidney, which may induce acute renal failure. Overdistention of the bladder can lead to permanent atony of the detrusor muscle and incontinence. Regardless, knowledge of a postoperative patient's urinary status is paramount, and the patient should be directly observed urinating within 6 hours postoperatively. In addition, the patient's bladder should be evacuated immediately before and after surgery, especially if an epidural has been administered.

Advice for Clients

Client education begins before surgery. In discussing postsurgical follow-up care including cage confinement and restricted activity, patient limitations should be made explicitly clear. Also advisement about certain forbidden activities (running, jumping, playing with other dogs, etc.) and the risk of implant failure or reinjury if the patient is not made to follow recommendations must be clearly explained. Despite the veterinary staff's best efforts these instructions may still be ignored. Providing written instructions for the owners is important for good follow through. One alternative approach is to remind owners that the surgical procedure is an investment, and if they follow postoperative care instructions they would be protecting their investment and avoiding costly treatment for complications.

Tricks of the Trade

The following are a few notes regarding specific situations or operations.

- TPLO patients may develop edema around the hock a few days postoperatively, usually after discharge. The client may call and express concern. The veterinary staff can explain that this pocket of fluid around the hock has developed as a result of the patient's lymphatic system being disturbed as well as a decrease in circulation from reduced limb use. The client can be instructed to perform gentle soft tissue massage to move the fluid up the leg and increase lymphatic return. The fluid pocket will resolve with time. In addition, warm compresses may help the edema to resolve sooner. If edema continues and there is any drainage from the incision, the patient should be evaluated for possible infection.
- Also regarding postoperative TPLO patients, occasionally a client may call expressing concern that their dog was walking very well, then suddenly became lame on the operative leg and now is sensitive around the surgical site. It is prudent to recheck these cases as soon as possible to evaluate for postoperative complications. Complications include, but are not limited to, infection, fracture of the proximal fibula, screw breakage or loosening of the screw, patella ligament swelling, fracture of the patella or tibial tuberosity, or rarely, fracture of the tibia.[23] Radiographs and a good orthopedic exam are essential in diagnosing complications. Fractures of the fibula, patella, or tibial tuberosity usually result from too much activity, which applies excessive forces on bones, combined with the acute changes in biomechanics following rotation of the tibial osteotomy. Fractures of the fibula heal with proper conservative management consisting of pain medication and crate confinement. Fractures of the tibia, screw breakage, or plate loosening generally require additional surgery, and treatment should be determined by the veterinary surgeon. Patella ligament swelling also occurs as a result of overuse before a period of tissue adaptation. Enforced rest and symptomatic treatment generally result in resolution of patella ligament swelling.
- Dogs that have surgery of the foot, carpus, or tarsus or distal limb fractures may have marked distal limb edema following surgery. The veterinary staff should be diligent in checking the toes to ensure that the bandage, if any is used, is not too tight. Distal limb edema can be attended to multiple times a day with superficial thermal modalities and massage, in conjunction with antiinflammatory medication and bandaging.
- In the immediate postoperative period following patella luxation repairs, care should be taken with passive range of motion exercises, particularly stifle flexion. In some cases, reluxation may occur if passive range of motion is too aggressive. In complex repairs of patella luxation, passive range of motion may be contraindicated in the first 2 weeks following surgery.

Nursing Care for the Neurologic Patient

Bladder Care

Urinary bladder management is a cornerstone of neurologic nursing care. The type of bladder management necessary depends on the location of the lesion and type of spinal injury. Bladder management in the neurologic patient plays an extremely important role in the overall recovery of the patient. Recurrent bladder infections can complicate the health of neurologic patients. Patients with intervertebral disk disease localized to the T3-L3 thoracolumbar region develop an upper motor neuron (UMN) bladder, whereas those with lower lumbar or sacral region lesions develop a (LMN) lower motor neuron bladder. Upper motor neuron bladders have a hypertonic bladder sphincter and are difficult to express. Lower motor neuron bladders have a very relaxed, hypotonic sphincter, are easily expressed, and are frequently incontinent. Both types of bladders may have overdistention of the bladder with resultant bladder damage if the bladder is not evacuated regularly.

Many neurologic patients are able to urinate voluntarily but choose not to. If they are unable to ambulate, they sometimes will not urinate on their own. An opportunity to urinate outside should always be given first. If a patient has been given the opportunity to urinate outside and is unsuccessful, the bladder must be emptied either by expression or catheterization.

It is extremely important to understand the difference between voluntary urination and bladder overflow and distention. Urine present in the patient's cage is not evidence that the patient can urinate independently. Knowing the difference is critical to proper bladder management. Bladder overdistention can eventually result in permanent atony of the detrusor muscle. The bladder must be evacuated frequently and kept to a small size to avoid this problem. If urine is noted in the cage of a neurologic patient, bladder palpation should be performed to ensure the bladder has been emptied. If the bladder remains large and palpable, and attempts to urinate outside were unsuccessful, there is likely overflow and bladder expression should be performed.

With any spinal injury, cystitis or bladder infections can become a serious threat to the patient's health. Clinical signs of a urinary tract infection include foul odor to the urine, bloody or dark-colored urine, flocculent or purulent material in the urine, urine dribbling, constant licking of the genital area, depression, loss of appetite, and, in advanced cases, fever.

Bladder Expression

There are many options for bladder palpation techniques and patient positioning. Ensuring the patient is relaxed minimizes stress involved with bladder expression. Placing the patient in a standing position or lateral recumbency helps the caregiver to palpate the bladder more easily.

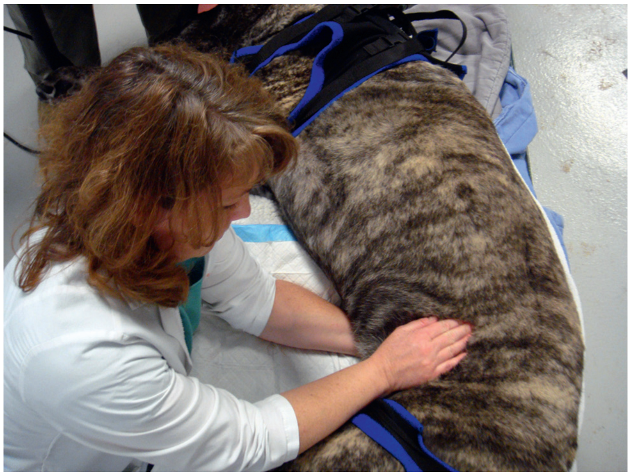

Figure 16-32 Bladder expression of the neurologic patient.

Place the hands under the abdomen between the last rib and the pelvis with one hand on either side of the patient. Slowly and gently palpate the patient's caudoventral abdomen for the presence of a bladder. Depending on the size of the bladder it may be located more cranially than one may anticipate. Place the hands cranially while applying gentle constant pressure to force the bladder to move caudally in between the caregiver's hands. It can be described as feeling somewhat like a water balloon that slips between your hands.

After the bladder is located, apply constant gentle pressure by compressing the two hands together, until urine is expressed. The bladder will gradually "deflate" as it empties (Figure 16-32). Bladder expression often can be extremely difficult, depending on the tenseness of the abdomen or if the patient is morbidly obese. Also, the difficulty to express may be caused by the underlying neurologic condition; with UMN bladders, spasm of the urethral sphincter may decrease or obstruct urine flow while expressing. Pharmacologic agents may be used to alter sphincter tone; the medication selected depends on whether bladder dysfunction is caused by UMN or LMN disease. Diazepam can be initiated if a UMN bladder is present to relax the skeletal portion of the urethral sphincter. This is especially important in male dogs. The initial dose of diazepam is 2 to 5 mg orally 30 minutes prior to bladder expression every 8 hours but can be increased as needed up to 10 mg. In addition, drugs may be used to relax the smooth muscle portion of the bladder, such as prazosin (1 mg/15 kg of body weight q 8 hours) or phenoxybenzamine (0.25 mg/kg po sid-bid, prn). Lower motor neuron bladders already have decreased sphincter tone, which makes it easy to express the bladder.

Bladder Catheterization

Most neurologic patients that have lost their ability to urinate can be managed with bladder expression alone. However, in some cases bladder catheterization is appropriate and often necessary. In a large recumbent patient, even if the ability to urinate is not lost, bladder catheterization may be necessary because of the inability to rise and move around. It must be noted that an increased risk for urinary tract infections exists in catheterized patients. The choice of whether to intermittently catheterize the patient or use an indwelling system depends on the type of urinary catheter, the experience and availability of the veterinary staff, and clinician preference.

The type of urinary catheter and length are important to properly evacuate and measure urine output. If intermittent catheterization (two to three times daily) is warranted, a red rubber or a soft polyvinyl catheter can be used. If a more long-term indwelling system is desired, a softer, less irritating catheter such as a Foley catheter with a balloon device for anchoring within the bladder can be used. Whichever system is used, sterile placement is important.

Materials Needed for Female Catheterization

- Scrub solution
- Tiemann or Foley catheter, appropriate sizes
- Sterile gloves
- Sterile lubricant
- Sterile speculum
- Water and syringe for balloon device
- Extension set
- Urine collection system and bag

Procedure for Female Catheterization

Two people are required for catheterization of a female dog. The dog is placed in sternal recumbency. Sedation is often necessary to decrease patient discomfort during catheter placement. The vulva is cleaned with scrub solution. Sterile gloves are worn and the urinary catheter and lubricant used are also sterile. Lubricant is placed on the catheter end. The sterile vaginal speculum should be introduced vertically through the vulva lips and then turned horizontally. Two small indentations are on the floor of the vestibule. One is near the clitoral fossa; the other is the urethral opening set on the urethral papilla. The catheter should be carefully introduced through the urethral papilla. After urine starts to flow, the balloon is inflated with water as indicated on the catheter, and attached to a urine collection system. Most female urinary collection systems are indwelling because of the difficulty associated with repeated cathterizations.

Materials Needed for Male Catheterization

- Antiseptic solution
- Sterile red rubber or Foley catheter
- Sterile gloves
- Sterile lube
- Extension set (if indwelling)
- Urine collection system and bag (if indwelling)
- Syringe, three-way stopcock, and bowl (short-term)

Procedure for Male Catheterization

Two people are required for the catheterization of a male dog. The dog is placed in lateral recumbency. The person restraining the dog extrudes the penis. The prepuce and penis are gently cleaned with antiseptic solution, such as dilute chlorhexidine. With the penis extruded, the person placing the catheter wears sterile gloves and is sterilely handed the urinary catheter and sterile lube. Sterile lube is placed over the catheter tip. The catheter is introduced into the tip of the penis and the catheter is gently advanced into the urethra until urine starts to flow. If an indwelling catheter system is required, a sterile extension set is attached to a urine collection bag. To ensure the catheter is anchored in the bladder, the balloon of the device is filled with water, with the amount dependent on the size of the catheter. The amount needed is noted on the neck of the catheter. If the catheter is temporary, a three-way stopcock may be attached to the end of the catheter and a 60-cc syringe may be used to empty or measure the urine (Figure 16-33).

Catheter Care

Indwelling urinary catheter systems should be cleaned with a mild disinfectant solution at the prepuce of males and the vulva of females. Check four times daily to be sure the catheter is in place and urine is continuing to flow into the collection bag. The urine bag should be placed on a clean surface below the patient, but not on the floor. Exam gloves should be worn when emptying the urinary collection bag. The extension set clamp is closed, and the exit valve on the collection bag is opened. Urine is collected in a graduated cylinder or bowl to be measured. Care should be taken to maintain sterility of the collection system as much as possible. Remember to open the clamp on the extension set when finished to allow urine to flow.

Special Considerations

Cervical Neurologic Patients

Atlantoaxial subluxation occurs when instability between the first two vertebrae causes a compression of the spinal cord. This disorder usually occurs in young toy or small breed dogs but can also be seen in larger breeds. Clinical signs can be of congenital origin or traumatic. The severity of the clinical signs will determine the course of treatment. Clinical signs can vary from mild ataxia to severe neck pain, tetraparesis, or tetraplegia. Conservative management for mildly affected cases includes strict cage confinement and brace placement for 4 to 6 weeks. The brace can be made from Thermoplast or fiberglass cast material, and extends from mid sternum cranially to the chin. The brace prevents the patient from ventral flexion of the neck (Figure 16-34). The brace should be closely monitored for excessive tightness, areas of irritation, and wetness. If any of these occur, the brace should be removed and replaced. Assure that the brace is not too tight, to allow the patient